1. Study of various Network Commands used in Linux and windows

1. arp - a

Show Ip address of your Computer along with Ip address and MAC address of your router.

output:

Interface 192.168.156.1 ... 0x7

Internet address physical address Type

192.168.56.255 ff-ff-ff-ff Static

224.00.22 01-00-5e-00-02

239.255.25.250 01-00-5e-ff-ff Static

2) hostname: Display name of your Computer

output:

blackie.

3) Ip Config /all:

Display detailed configuration information about TCP/Ip Connection and type of ethernet adapter in the system.

output:

window Ip configuration.

Host name : blackie

primary DNS Suffix .

Node type.

4) nbstat -a:

Display protocol statistics and current, TCP/IP connection on using NBT

output:

NBTSTAT [[-a Remote Name] [A IP Address] [-]

5) netstat:

Display a variety of statistics about a Computer, active Tcp/Ip Connection

output:

Active Connection

proto	Local address	Foreign address	state
Tcp	10.76.96.60.4915	10:76.96.163.do	Timeo
Tcp	10.76.96.60.52399	142.250.183.bp	ESTABLISH
Tcp	10.76.96.60.56954	10.76.96.163	TIME_WAIT

6) nslookup

Display DNS detail such as Ip address and the particular Comperter

output:

Server: unknown

Address: 10.76.196.163

pathping: Traces the root to destination address the launches a 25 second. test of each router along the way.

output:

pathping [-ghost-hust] [-h maximum-hops] [-i address]

Linux Networking Commands

1) ip: help form setting up new System and assigning Ip la trouble Shooting exusting System.

2) ifconfig: The ifconfig is staple in many Sys admin's tool fellt for Confeguring trouble shooting. networks

3) mtr: TNTR is program with CLI that are a network diagnostic and throws trouble shooting tool.

4) Pong: ping is a tool that verifies IP level Connectivity to another TCP/IP Computer by sending Internet Control Message protocol (ICMP) Echo Request message. The receipt of corresponding Echo Reply message protocol frame and reachability resolution

Configuring an Ethernet connection by using nmcli.

if you cannot a host to the network over Ethernet, you can manage the connections setting on the command line by using the utility

procedure:

1) List the Network Manager connection profiles.
nmcli connection show.

2) nmcli connection add con-name <Connection name> ifname <device-name> type ethernet

skip this step. to modify an existing profile

3) optional: Rename the Connection profile:

nmcli connection modify "wired Connection 1"

Here, "wired connection 1" is the name of the Connection.

4) Display the current setting of the Connection profile

nmcli Connection show.

5) Configure the IPv4 Settings:-

To use DHCP, enter.

nmcli connection modify "wired Connection 1"

ipv4 meathod auto.

6) Configure the Ipv6 Settings:

#nmcli connection modify "wired Connection 1" ipv6. meathod auto.

7. Activate the profile

#nmcli connection up Internal - LAN.

Result:

Therefore, the various of Network comands are executed successfully.